

A Technical Introduction to Linear regression

$$\hat{y} = X\beta + \varepsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Section 1

$$\xi(w) = w_1 \beta_1 + w_2 \beta_2 + \dots + w_q \beta_q, \quad \{\text{displaystyle } \xi(w) = w_1 \beta_1 + w_2 \beta_2 + \dots + w_q \beta_q\}$$

Section 2

General linear models The general linear model considers the situation when the response variable is not a scalar (for each observation) but a vector, y_i . Conditional linearity of $E(y | x) = x^T B$ is still assumed, with a matrix B replacing the vector β of the classical linear regression model. Multivariate analogues of ordinary least squares (OLS) and generalized least squares (GLS) have been developed. "General linear models" are also called "multivariate linear models". These are not the same as multivariable linear models (also called "multiple linear models").

Section 3

$Y = \ln \beta_1 X + \epsilon$ In the above equations, the coefficients (parameters) are non-linear. A model with exactly one explanatory variable is a simple linear regression; a model with two or more explanatory variables is a multiple linear regression. This term is distinct from multivariate linear regression, which predicts multiple correlated dependent variables rather than a single dependent variable. In linear regression, the relationships are modeled using linear predictor functions whose unknown model parameters are estimated from the data. Most commonly, the conditional mean of the response given the values of the explanatory variables (or predictors) is assumed to be an affine function of those values; less commonly, the conditional median or some other quantile is used. Like all forms of regression analysis, linear regression focuses on the conditional probability distribution of the response given the values of the predictors, rather than on the joint probability distribution of all of these variables, which is the domain of multivariate analysis. Linear regression is also a type of machine learning algorithm, more specifically a supervised algorithm, that learns from the labelled datasets and maps

the data points to the most optimized linear functions that can be used for prediction on new datasets. Linear regression was the first type of regression analysis to be studied rigorously, and to be used extensively in practical applications. This is because models which depend linearly on their unknown parameters are easier to fit than models which are non-linearly related to their parameters and because the statistical properties of the resulting estimators are easier to determine. Linear regression has many practical uses. Most applications fall into one of the following two broad categories:

Section 4

parameter β_j of predictor variable x_j represents the individual effect of x_j . It has an interpretation as the expected change in the response variable y when x_j increases by one unit with other predictor variables held constant. When x_j is strongly correlated with other predictor variables, it is improbable that x_j can increase by one unit with other variables held constant. In this case, the interpretation of β_j becomes problematic as it is based on an improbable condition, and the effect of x_j cannot be evaluated in isolation. For a group of predictor variables, say, $\{x_1, x_2, \dots, x_q\}$, a group effect $\xi(w)$ is defined as a linear combination of their parameters

Section 5

$$X = [x_1^T \ x_2^T \ \dots \ x_n^T] = [1 \times 11 \ \dots \ 1 \times 21 \ \dots \ 1 \times n1 \ \dots \ 1 \times np], \quad \{\text{displaystyle } X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}\}$$

Section 6

Note: The $\hat{\beta}$ obtained may indeed be the local minimum, one needs to differentiate once more to obtain the Hessian matrix and show that it is positive definite. This is provided by the Gauss–Markov theorem. Linear least squares methods include mainly:

Section 7

$$l(D, \beta) = \log \prod_{i=1}^n \Pr(y_i | x_i \rightarrow; \beta, \sigma) = \log \prod_{i=1}^n \frac{1}{2\pi\sigma} \exp\left(-\frac{(y_i - \beta \cdot x_i)^2}{2\sigma^2}\right) = n \log \frac{1}{2\pi\sigma} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta \cdot x_i)^2 \quad \{\text{displaystyle } l(D, \vec{\beta}) = \log \prod_{i=1}^n \Pr(y_i | \vec{x}_i, \vec{\beta}, \sigma) = \log \prod_{i=1}^n \left(\frac{1}{\sqrt{2\pi}\sigma}\right) \exp\left(-\frac{(y_i - \vec{\beta} \cdot \vec{x}_i)^2}{2\sigma^2}\right)\}$$

$$\frac{1}{n} \log \left(\frac{1}{\sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ij} \right)^2} \right) = -\frac{1}{2n} \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ij} \right)^2$$

Section 8

Errors-in-variables models (or "measurement error models") extend the traditional linear regression model to allow the predictor variables X to be observed with error. This error causes standard estimators of β to become biased. Generally, the form of bias is an attenuation, meaning that the effects are biased toward zero.

Section 9

$$Y_{ij} = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} + \epsilon_{ij}$$

Section 10

Maximum likelihood estimation Maximum likelihood estimation can be performed when the distribution of the error terms is known to belong to a certain parametric family f_θ of probability distributions. When f_θ is a normal distribution with zero mean and variance θ , the resulting estimate is identical to the OLS estimate. GLS estimates are maximum likelihood estimates when ϵ follows a multivariate normal distribution with a known covariance matrix. Let's denote each data point by (x_i, y_i) and the regression parameters as β , and the set of all data by D and the cost function by $L(D, \beta) = \sum_{i=1}^n (y_i - \beta \cdot x_i)^2$. As shown below the same optimal parameter that minimizes $L(D, \beta)$ achieves maximum likelihood too. Here the assumption is that the dependent variable y is a random variable that follows a Gaussian distribution, where the standard deviation is fixed and the mean is a linear combination of x :

$$H(D, \beta) = \prod_{i=1}^n \Pr(y_i | x_i; \beta, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(y_i - \beta \cdot x_i)^2}{2\sigma^2} \right)$$

$$\begin{aligned} H(D, \beta) &= \prod_{i=1}^n \Pr(y_i | x_i, \beta, \sigma) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(y_i - \beta \cdot x_i)^2}{2\sigma^2} \right) \end{aligned}$$